

Australia to Japan energy supply chain analysis

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1 Introduction

1.1 Motivation

Energy prices play a pivotal role in a nation's economy and market. Understanding how price changes move through the energy supply chain is essential in designing energy policies, managing trade, and forecasting impact. Steaming coal remains a major energy source for generating electricity in Asia, where coal prices influences energy costs, production, trade, and climate policy decisions. Australia is the fifth-largest producer, and second-largest exporter, of coal. Its reserves, mining sector, and port infrastructure have made it a leading global supplier. Japan, in contrast, is one of the largest coal importers, with limited domestic energy resources and a high dependence on imported fossil fuels. Its reliance on Australia's coal exports, ranked Japan first in major export markets for Australia in 2018. This strong reliance means that changes in Australia's export prices transmit directly to Japan's import costs and demand for coal.

1.2 Research Questions

This proposal focuses on the following research questions:

1. What is the overall trend for all three series, and what is the long-run relationship between them?
2. How accurately can Australia's coal export prices and Japan's import costs be forecasted?
3. To what extent, and over what horizon, do shocks to Australia's coal export prices effect Japan's import costs and import volume? How quickly are deviations from the long-run horizon corrected?

To investigate these questions, we will focus on three particular time series.

- Australia's Export price of Steaming coal to Japan.
- Japan's Import cost of Steaming coal from Australia.
- Japan's coal import volume from Australia.

2 Data selection and Exploration

2.1 Dataset selection

We will be using two datasets provided by Borealis: The Canadian Dataverse repository.

1. IEA Energy Prices and Taxes Statistics. EU, Japan - Coal import costs and volumes by origin.
2. IEA Energy Prices and Taxes Statistics. Australia, Canada, US - Coal export prices by destination.

The first one contains Japan's steaming coal import cost and import volume from Australia, and the second one contains steaming coal price exports from Australia to Japan

2.2 Exploration

2.2.1 Characteristics

From Figure 5, the time series starts from 1980 Q1 to 2011 Q2, the values are reported quarterly, and there are no missing values between the start and end dates.

There is an outlier in the last observation of the import volume that has to be dealt with. The import volume is unusually high at 31403.34 and is the last observation, this is likely a mis-input as there was no major global event or source that would have caused such a spike in the import volume. The last value was forecasted using a 1-step ahead predictor from an ARIMA(1, 1, 3) with drift¹ in the original series. The summary statistics of the three series can be seen below in Table 1.

Figure 6 (Appendix) shows there is high correlation between the import price and export cost of coal from Australia to Japan, and moderate correlation between import volume to both export and import cost.

¹This was chosen via. `auto.arima()`

Series AUS to JPN	Mean	Variance	Min	Q1	Median	Q3	Max
Export price US\$/tonne	43.717	425.495	24.11	32.5	38.255	45.598	124.78
Import price US\$/tonne	56.753	608.604	33.702	42.404	49.635	61.186	159.051
Import volume 1000 tonne	8549.311	27504533	462.04	4489.713	7851.826	12911.14	19357.14

Table 1: Summary Statistics for Australia to Japan Steaming Coal Export/Import/Demand

2.2.2 Time Series Plots

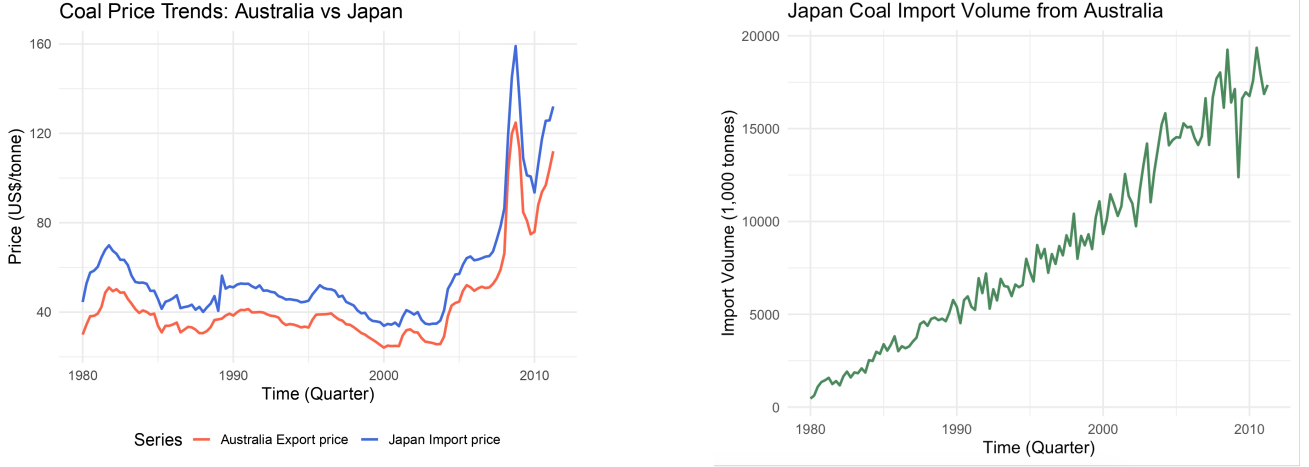


Figure 1: Time Series Plot for Prices (Left) and Volume (Right)

Figure 1 displays quarterly trends in coal prices and import volumes between Australia and Japan from 1980 to 2011. All three series show an upward trend. The left panel shows a tight price linkage between Australian export prices and Japanese import prices, and the right shows a steady increase in import volumes.

From Figure 7 and 8 (Appendix), all three series exhibit random-walk behavior. The ACF of the three series show a slow decay and strong persistence across many lags. The PACF shows a dominant lag-1 spike followed by rapid decay. These are all indicators of non-stationarity, and differencing will be required to achieve stationarity.

The decomposition shown in Figure 10 (Appendix) show there is mostly a trend component, which is consistent with what was seen in the ACF, showing slow decay and non-stationarity. There is a small seasonal and random component, so quarterly fluctuations are minor compared to the underlying trend.

2.3 Transformations

The ACFs and PACFs of the first order difference observed in Figure 11 (Appendix) and the Augmented Dickey-Fuller tests observed in Figures 12 to 18 (Appendix) indicate that the first-order differences are stationary. The CCF between import and export prices indicate strong positive cross correlation at lag 0. This is expected since export and import changes should move simultaneously. Volume and import/export prices exhibit weak cross correlation.

3 Analysis, Implementation, and Results

3.1 Model selection

From the initial exploration, Australian export prices and Japanese import coal prices move together over time, and are non-stationary. Unit root tests indicate that both are integrated of order 1, $I(1)$, while their differences are stationary and $I(0)$. The Johansen tests and Engle-Granger procedure 16 confirm there is the presence of at least 1 cointegrating relationship, suggesting that the series move together in the long run. Given that we have non-stationary levels, stationary differences, and cointegration, a Vector Error Correction Model would be the appropriate model to fit. This also ties in well to our research questions of studying impulse responses, forecasting, and how the two series co-move over time.

3.2 Model Parameter choice

Doing a VAR test for the three series shown in Table 4 (Appendix) suggests $p = 6$ with AIC and FPE criterion, $p = 5$ for HQ criterion, and 4 for BIC criterion. AIC and FPE tend to overfit and like more lags where BIC penalizes expressive models. A 3-variable VAR(6) model will have 54 parameters, and is likely too complex. We will fit with $p = 5$, and a total of 45 parameters. The Johansen test results presented in Figure 19 (Appendix) indicates cointegration, with our test rejecting both null hypothesis $H_0 : \text{rank} \leq 0$ and $H_0 : \text{rank} \leq 1$, and failing to reject null hypothesis $H_0 : \text{rank} \leq 2$ at the 5% level of significance. It's worth noting that at the 1% level, the test rejects $H_0 : \text{rank} \leq 0$, however, it fails to reject the null hypothesis $H_0 : \text{rank} \leq 1$. The choice between adding 1 or 2 cointegrating equations is borderline, and since one of our research questions deals with interpreting the long run horizon, we decided to fit with one cointegrating equation rather than 2. Interpretation of one cointegrating equation is much more elementary than the interpretation of two cointegrating equations.

3.3 Model Results

Based on the fitted model presented in Figure 20, the estimated long-run cointegrating relationship is

$$\text{ECT}_{t-1} = \text{import_price}_{t-1} - 1.2417\text{export_price}_{t-1} + 0.000232\text{import_volume}_{t-1} - 1.558$$

The magnitude of the export price(+1.2417) is dominating, and tells us that Australia's export price strongly anchors Japan's import price. The small magnitude of the import volume tells us that volume has a small role in determining long-run price level. The short-run equations from Figure 20 is

$$\Delta \text{import_price}_t = 0.1495\text{ECT}_{t-1} + \dots$$

$$\Delta \text{export_price}_t = 0.3259\text{ECT}_{t-1} + \dots$$

$$\Delta \text{import_volume}_t = 115.9\text{ECT}_{t-1} + \dots$$

Based on the error-correction dynamics, prices deviate, Japan's import price adjusts upward moderately, so this has slow correction. Australia's export price adjusts faster than Japan's, so corrections originate more strongly in Australia. Volume reacts strongly to disequilibrium, and correct themselves. So when prices mismatch, trade quantities adjust very quickly.

$$\text{Letting, } \alpha = \begin{pmatrix} 0.1495 \\ 0.3259 \\ 115.9 \end{pmatrix} \text{ and } \Delta y_t = \begin{pmatrix} \Delta \text{import_price}_t \\ \Delta \text{export_price}_t \\ \Delta \text{import_volume}_t \end{pmatrix}$$

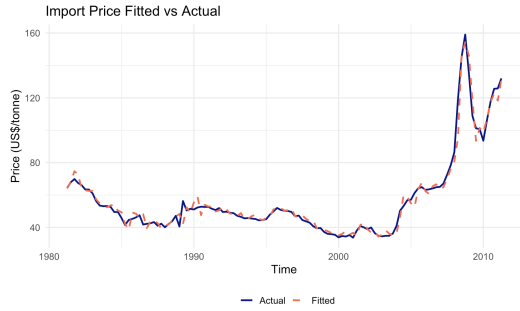
The VECM model can be written as:

$$\Delta y_t = \alpha \text{ECT}_{t-1} + \Gamma_1 \Delta y_{t-1} + \Gamma_2 \Delta y_{t-2} + \Gamma_3 \Delta y_{t-3} + \Gamma_4 \Delta y_{t-4}$$

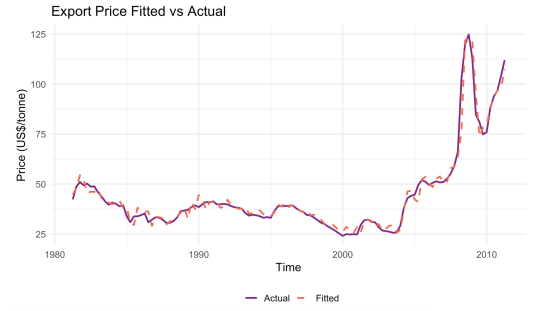
$$\begin{aligned} \Gamma_1 &= \begin{pmatrix} -0.2816 & 0.9571 & 0.0001858 \\ 0.2897 & 0.2823 & -0.0005586 \\ -66.24 & 151.80 & -0.7103 \end{pmatrix} & \Gamma_2 &= \begin{pmatrix} -0.03832 & 0.07777 & 0.0004838 \\ 0.3722 & -0.4647 & 0.0002813 \\ -57.93 & -23.83 & -0.5834 \end{pmatrix} \\ \Gamma_3 &= \begin{pmatrix} 0.5266 & -0.8473 & 0.0007006 \\ 0.7387 & -1.0040 & 0.0003309 \\ 22.09 & 16.63 & -0.5327 \end{pmatrix} & \Gamma_4 &= \begin{pmatrix} 0.7876 & -1.0260 & 0.000004549 \\ 0.6130 & -0.8924 & -0.0004922 \\ 11.72 & -99.57 & 0.04136 \end{pmatrix} \end{aligned}$$

There are few things we can interpret from the lag structure, lagged changes in Australia export prices have strong persistence on Japan's import prices, especially, at smaller lags. Import volumes show their own strong lag persistence, with negative coefficients on the first two lagged values.

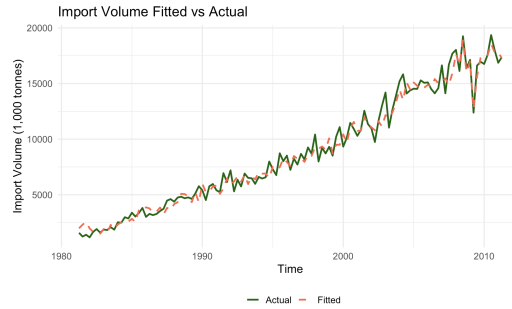
3.3.1 Fitted Model



(a) Fitted Model for Import Prices



(b) Fitted Model for Export Prices

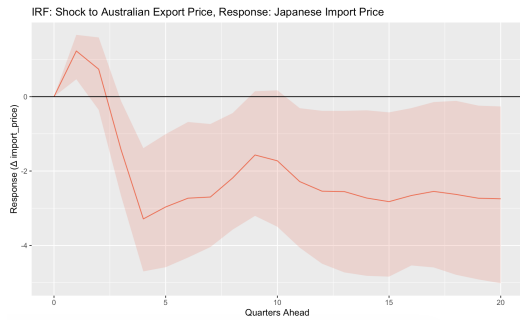


(c) Fitted Model for Import Volume

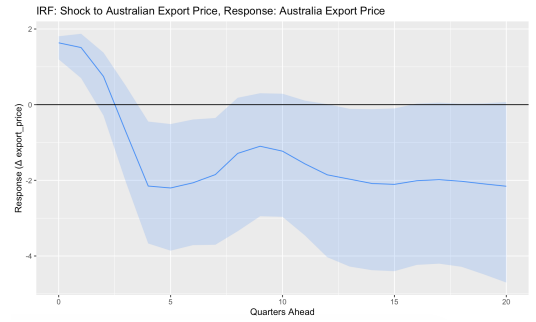
Figure 2: Fitted Model Plots

3.4 Impulse Response Function

Impulse Response Function (IRF) shows how one variable responds over time to a shock (1-unit innovation) in another variable, keeping everything else constant. A natural question for us to ask is, "if we shock Australia's export price, then how will the other variables move?"



(a) Import price response



(b) Export price response



(c) Import volume response

Figure 3: IRF plots when shocking Australian Export prices

When shocking the export price, initially, the import price is raised, but immediately spikes downwards, turns negative and remains negative in the long run. The confidence bands being below 0 show the persistent

adjustment. On the export price, the price is initially raised which is as expected, then drop below 0, but slowly converges back to 0. The export price is closer to 0 than the import price in the long run. The import volume bears the most adjustment, there is an immediate spike down, and the imports remain consistently low when prices jump. These results are consistent with the ECT coefficients seen earlier.

To check reverse causality, we shock the import price and see the effects. The IRF plots when shocking Japanese import prices can be seen in Figure 21 in the Appendix. Although the shocks show both the import price and export price growing, it returns to the baseline value ~ 4 . However, since the baseline doesn't return to 0, the shocks still cause a non-zero deviation over time.

3.5 Model Diagnostics

3.5.1 Testing Autocorrelation

To test the autocorrelation of the residuals, we will use the Portmanteau Test for VECM residuals. This is similar to Ljung-box test, and the results are seen in Figure 22 in the Appendix. A p-value of 0.1714 indicates that there is no residual autocorrelation for up to 16 lags, and capturing the correlation structure properly. The ACF and PACF of the residuals observed in Figures 25 and 26 in the Appendix confirms this, as the correlation disappears rapidly in the ACF, and the correlation in the PACF is non-significant.

3.5.2 Testing Normality

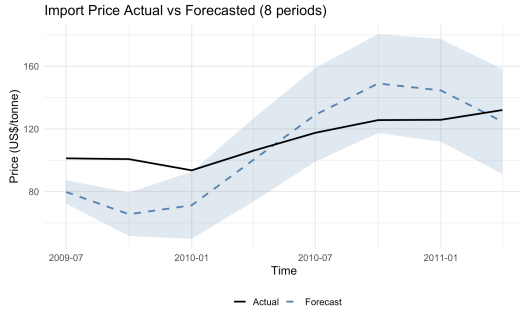
The Jarque-Bera test for normality observed in Figure 23 in the Appendix indicate a strong rejection of normality. The spikes seen in Figure 24 in the Appendix of the residual plots likely cause rejection of multivariate normality. Even after log-transforming all the variables, we still have strong rejection from the test seen in Figure 27 (Appendix). However, according to one such paper, Johansen's procedure is robust to non-normality [1], and forecasting won't be affected by non-normality either.

3.5.3 ARCH test

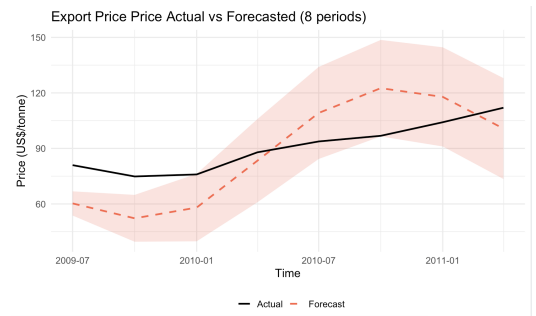
The ARCH test seen in Figure 28 in the Appendix suggests that we reject the null hypothesis that we have no multivariate ARCH effects, and we have heteroskedastic residuals for lagged values of 5. Taking log of all time series also rejects the null hypothesis at $\alpha = 0.05$ observed in Figure 29 in the Appendix.

3.6 Forecasting Results

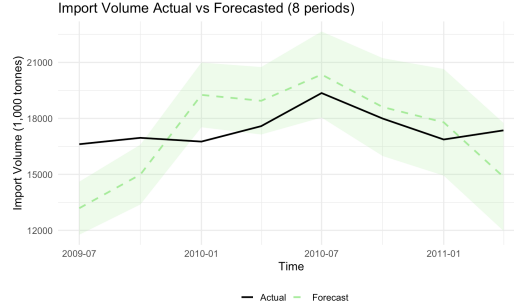
Removing the last 8 observations (2 years) of our dataset, we forecast the next 8 observations and compare that with the actual observed points. Figure 4 suggests that the model was slightly more expressive than the actual data, but most of the points were contained in the 95% confidence interval for prices. The model didn't perform as well when predicting the import volume. The RMSE, MAE, and MAPE for the three series can be seen in Table 2.



(a) Import Price Forecast vs. Actual for last 8 periods



(b) Export Price Forecast vs. Actual for last 8 periods



(c) Import Volume Forecast vs. Actual for last 8 periods

Figure 4: Forecast plots

We want to compare the fitted VECM model to univariate SARIMA models to see if the multivariate approach improves predictive accuracy and if including all series is necessary for forecasting. Using stepwise model selection for each series individually, and fitting based off of AIC. For import price, the fitted model is $ARIMA(3, 1, 0)$, for export price, the fitted model is $ARIMA(2, 1, 1)$, and for import volume, the fitted model is $ARIMA(0, 1, 1)$. See Figure 30.

	Import Price	Export Price	Import Volume
RMSE	20.35684	17.66126	2005.077
MAE	18.23125	16.48291	1784.85
MAPE	18.84242%	16.80606%	10.40801%

Table 2: RMSE, MAE, and MAPE for the 8 forecasted values for VECM model

	Import Price	Export Price	Import Volume
RMSE	13.00936	30.14817	349.3726
MAE	10.15676	29.15717	571.8239
MAPE	9.078022%	33.30406%	3.126189%

Table 3: RMSE, MAE, and MAPE for the 8 forecasted values using ARIMA models

We see that the SARIMA models performed much better on the import price, and import volume forecasting. However, it was much worse at predicting export prices. Surprisingly, the fitted models are slightly different for the import price and export price series, even though they are very similar. It seems like fitting an MA component for export price performed much worse compared to the $ARIMA(3, 1, 0)$ of the Import Price. When fitting the export series with the $ARIMA(3, 1, 0)$ model, we still get forecasting metrics worse than the VECM model but perform better than the $ARIMA(2, 1, 1)$ model.

4 Conclusion

4.1 RQ1:

The overall long-term trend in coal prices and import volumes behaves like a Random Walk based on the plot of the original series, and the non-stationary behavior from the ACF/PACF plots and ADF tests. The drift

term in the ARIMA model fitted for import price is 0.7032. This indicates that the Japanese coal import prices rise by approximately \$0.732 per tonne per quarter on average in the long run. The drift term in the ARIMA model fitted of 0.7214 for export price is similar to the import price. Export prices rise approximately by \$0.7214 per tonne per quarter on average in the long run. Finally, the drift term in the ARIMA model fitted for import volume is 138.0583. Import volume increases by about 138.0583 tonnes per quarter, on average, in the long run. The long-run equilibrium equation from the VECM model shows the tied relationship between import and export prices. Specifically, the dominating coefficient of the export price shows that it anchors the import price. The small magnitude of the import volume plays a small role for the long-run price level.

4.2 RQ2:

Although the VECM model is able to capture the long-run equilibrium dynamics between the coal prices and import volumes, the predictive power was much smaller compared to the individual ARIMA models. The VECM model only performed better on export prices compared to the individual ARIMA models when forecasting the export price. The individual ARIMA models performed much better on the import price, and the import volume series. So, the multivariate approach doesn't necessarily improve predictive accuracy and isn't necessarily needed for predictive forecasting. Even though the ARIMA performs worse on the export price, when using an ARIMA(3, 1, 0) rather than ARIMA(2, 1, 1), the performance is closer to the VECM model. The VECM model perhaps could perform better by including a 4th series that captures overall global energy prices such as oil, or global prices, additionally, including the extra cointegrating relationship might be beneficial for predicting as well.

4.3 RQ3:

The impulse response functions show a positive shock to Australian export prices initially increases Japanese import prices, indicating short-run pass-through. However, we see an immediate downward spike, indicating the response quickly reverses and remains negative over the long term. The confidence bands being below 0 indicate the persistent adjustment. The export price also has an initial rise, but quickly drops below 0, but doesn't drop as far as the Japanese import prices. In the long run, the export price is closer to 0 than the import price. The import response functions for the import volume are consistently negative, which shows that import volumes react strongly and persistently. Overall, this highlights Japan's limited price-setting power compared to Australia, and will decrease the amount of imports in response to price shocks. These dynamics are consistent with the VECM coefficients that were seen earlier. The coefficients of the error-correction term are the highest in the import volume, and the export price coefficient is larger than the import prices. The relatively small loading on the import price coefficient implies that deviations from the long-run relation are only weakly corrected though changes in the import price. Finally, the very large coefficient of the import volume, combined with the large negative lag coefficients in, $\Delta \text{import_volume}_t$ reflects consistently with the observation in the IRF plot for import volumes. The import volume quantity reacts very sharply, and remain depressed, and indicating that the burden of adjustment falls on the import volumes rather than the prices.

4.4 Future Directions

It's important to note that coal is a globally traded commodity, and Australia itself has little control over the export price of coal. The model can be further improved by including a global coal price index series, and Australia's coal export volume. Global coal price would likely be a strong indicator of the export price, and Australia's coal export volume could have an effect on the global price index. Including these series could have stronger implications and interpretations for the Vector Error Correction Model as well. Our series also only went to 2011, finding a series with more recent points will mean more data for our models to fit. This would be beneficial to models like VECM that require a large amount of data to fit many parameters. Finally, since we are looking at price series, we can obtain returns, and utilize GARCH and volatility modeling can be used to estimate the variance of the series.

5 Appendix

	series	start	end		n_total	n_missing	n_obs	pct_missing
1	Import Price	1980 Q1	2011 Q2		126	0	126	0
2	Import Volume	1980 Q1	2011 Q2		126	0	126	0
3	Export Price	1980 Q1	2011 Q2		126	0	126	0

Figure 5: Time series completeness summary for Import and Export indices, 1980 Q1–2011 Q2.

	import_price	import_volume	export_cost
import_price	1.0000000	0.5344057	0.9935590
import_volume	0.5344057	1.0000000	0.5711171
export_cost	0.9935590	0.5711171	1.0000000

Figure 6: Correlation matrix of the three series

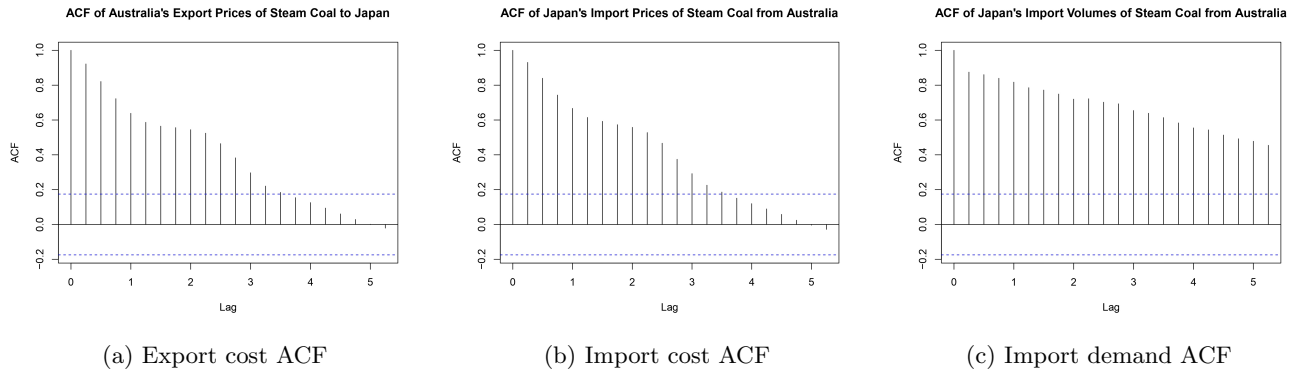


Figure 7: ACF plots

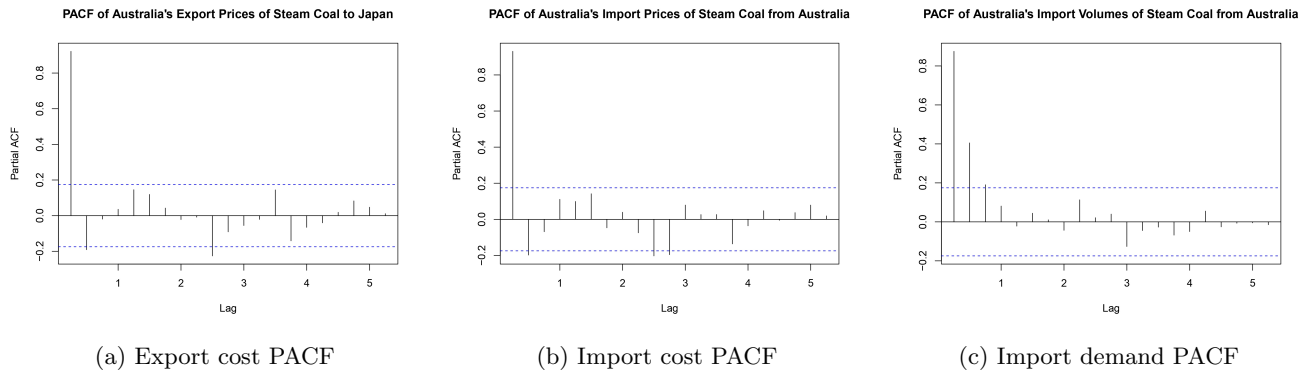
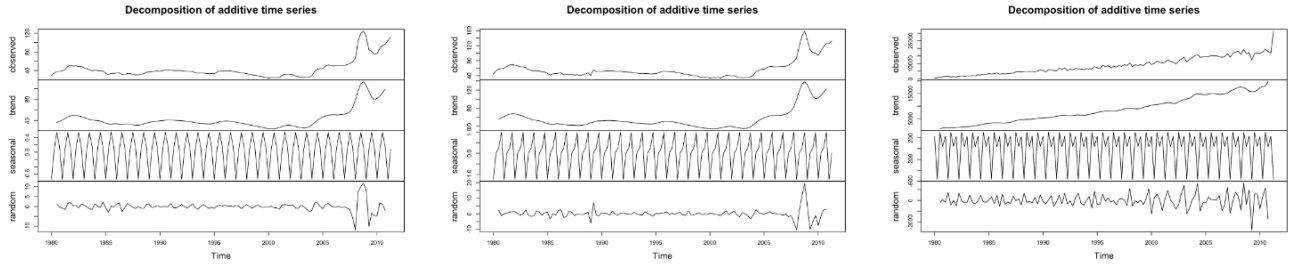


Figure 8: PACF plots



(a) Export cost decomposition

(b) Import cost decomposition

(c) Import demand decomposition

Figure 9: Decomposition plots

Component	% SS
Trend	99.046544400
Seasonality	0.009799375
Random	0.300150888

Export cost

Component	% SS
Trend	98.99809849
Seasonality	0.01735591
Random	0.24927819

Import cost

Component	% SS
Trend	99.7308640
Seasonality	0.1162352
Random	0.3990905

Import demand

Figure 10: Percentage of Variance Explained (%SS).

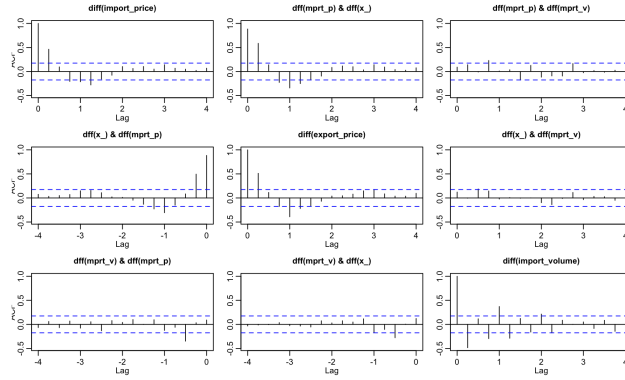


Figure 11: ACF and CCF of differenced series

Augmented Dickey-Fuller Test

```
data: data$import_price
Dickey-Fuller = -0.76805, Lag order = 4, p-value = 0.9621
alternative hypothesis: stationary
```

Figure 12: ADF test for Import prices

Augmented Dickey-Fuller Test

```
data: data$export_cost
Dickey-Fuller = -0.099338, Lag order = 4, p-value = 0.99
alternative hypothesis: stationary
```

Figure 13: ADF test for Export prices

Augmented Dickey-Fuller Test

```
data: diff(import_price)
Dickey-Fuller = -6.8638, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary
```

Figure 14: ADF test for 1st order differencing on import price

Augmented Dickey-Fuller Test

```
data: diff(export_price)
Dickey-Fuller = -5.1225, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary
```

Figure 15: ADF test for 1st order differencing on export price

Augmented Dickey-Fuller Test

```
data: residuals(eg_reg)
Dickey-Fuller = -4.4041, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary
```

Figure 16: Engle-Granger test on residuals of import price $\sim$ export price

Augmented Dickey-Fuller Test

```
data: data$import_volume
Dickey-Fuller = -3.1562, Lag order = 4, p-value = 0.0984
alternative hypothesis: stationary
```

Figure 17: ADF test on import volumes

Augmented Dickey-Fuller Test

```
data: diff(data$import_volume)
Dickey-Fuller = -7.2077, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary
```

Figure 18: ADF test for first order differencing on import volumes

\$selection
AIC(n) HQ(n) SC(n) FPE(n)
6 5 4 6

	1	2	3	4	5	6	7	8	9	10
AIC(n)	19.03	18.21	17.92	17.59	17.41	17.38	17.40	17.43	17.54	17.57
HQ(n)	19.15	18.41	18.20	17.96	17.87	17.93	18.03	18.15	18.35	18.47
SC(n)	19.31	18.71	18.63	18.51	18.55	18.74	18.96	19.21	19.53	19.78
FPE(n)	1.84×10^8	8.12×10^7	6.04×10^7	4.36×10^7	3.64×10^7	3.58×10^7	3.64×10^7	3.79×10^7	4.25×10^7	4.47×10^7

Table 4: VAR select for 3 variable TS

```
#####
# Johansen-Procedure #
#####

Test type: maximal eigenvalue statistic (lambda max) ,
without linear trend and constant in cointegration

Eigenvalues (lambda):
[1] 2.814439e-01 1.351769e-01 5.686491e-02 -9.515587e-17

Values of teststatistic and critical values of test:

      test 10pct  5pct  1pct
r <= 2 |   7.08   7.52   9.24 12.97
r <= 1 |  17.57  13.75  15.67 20.20
r = 0  |  39.99  19.77  22.00 26.81

Eigenvectors, normalised to first column:
(These are the cointegration relations)

      import_price.l1 export_price.l1 import_volume.l1
import_price.l1      1.0000000000      1.000000e+00      1.0000000000
export_price.l1     -1.2416580590     -1.186016e+00     -1.300597338
import_volume.l1      0.0002317032     -4.834202e-05      0.001035992
constant           -1.5582012369     -5.611734e+00     -9.468589071

      constant
import_price.l1      1.0000000000
export_price.l1     -1.7247018328
import_volume.l1      0.0005195379
constant             13.1520235287
```

Figure 19: Johansen Procedure for import, export prices, and import volume

```
$rlm
```

```
Call:
```

```
lm(formula = substitute(form1), data = data.mat)
```

```
Coefficients:
```

	import_price.d	export_price.d	import_volume.d
ect1	1.495e-01	3.259e-01	1.159e+02
import_price.dl1	-2.816e-01	2.897e-01	-6.624e+01
export_price.dl1	9.571e-01	2.823e-01	1.518e+02
import_volume.dl1	1.858e-04	-5.586e-04	-7.103e-01
import_price.dl2	-3.832e-02	3.722e-01	-5.793e+01
export_price.dl2	7.777e-02	-4.647e-01	-2.383e+01
import_volume.dl2	4.838e-04	2.813e-04	-5.834e-01
import_price.dl3	5.266e-01	7.387e-01	2.209e+01
export_price.dl3	-8.473e-01	-1.004e+00	1.663e+01
import_volume.dl3	7.006e-04	3.309e-04	-5.327e-01
import_price.dl4	7.876e-01	6.130e-01	1.172e+01
export_price.dl4	-1.026e+00	-8.924e-01	-9.957e+01
import_volume.dl4	4.549e-06	-4.922e-04	4.136e-02

```
$beta
```

	ect1
import_price.l1	1.0000000000
export_price.l1	-1.2416580590
import_volume.l1	0.0002317032
constant	-1.5582012369

Figure 20: VECM model for export, import prices and import volume

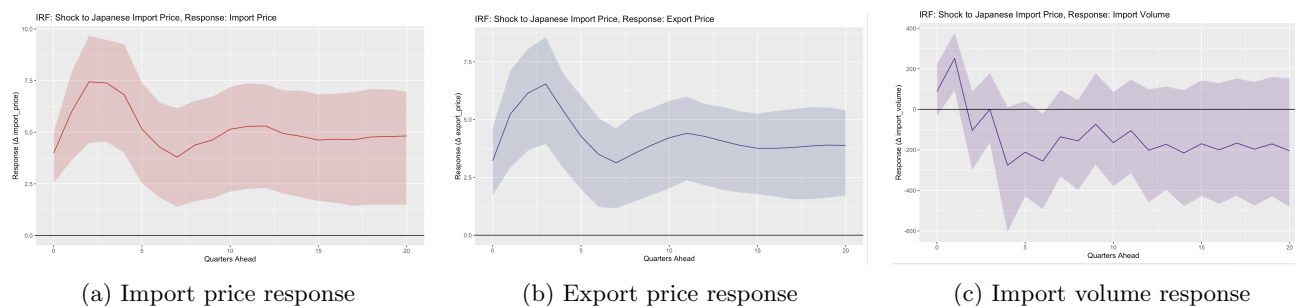


Figure 21: IRF plots shocking import price

Portmanteau Test (asymptotic)

```
data: Residuals of VAR object var3
Chi-squared = 115.44, df = 102, p-value = 0.1714
```

Figure 22: Portmanteau Test for autocorrelation of residuals

\$JB

JB-Test (multivariate)

data: Residuals of VAR object var3
Chi-squared = 819.68, df = 6, p-value < 2.2e-16

\$Skewness

Skewness only (multivariate)

data: Residuals of VAR object var3
Chi-squared = 90.783, df = 3, p-value < 2.2e-16

\$Kurtosis

Kurtosis only (multivariate)

data: Residuals of VAR object var3
Chi-squared = 728.9, df = 3, p-value < 2.2e-16

Figure 23: Jarque-Bera test for normality

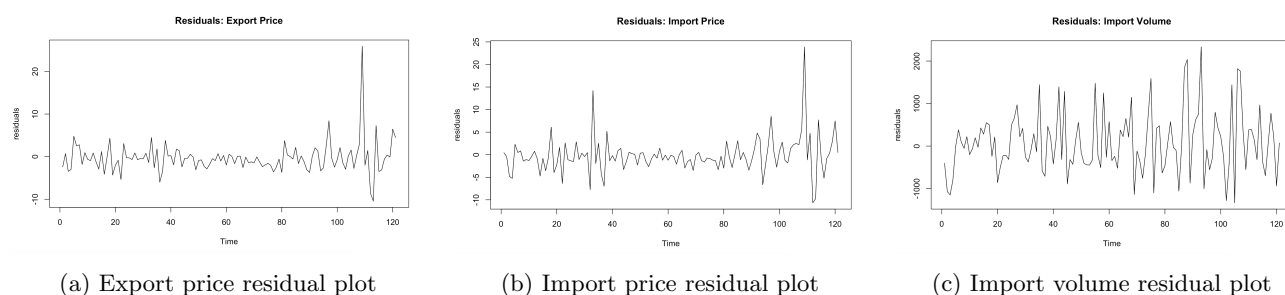


Figure 24: Residual Plots

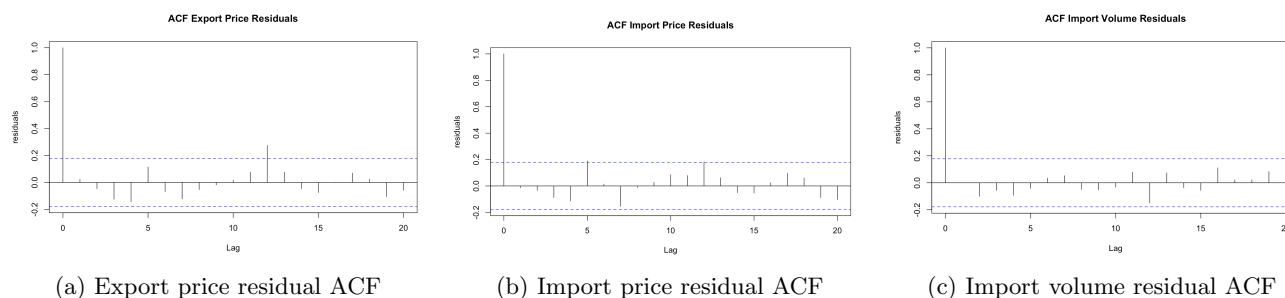
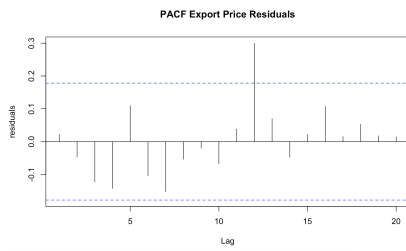
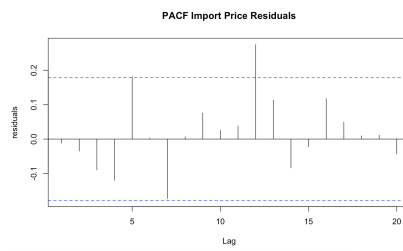


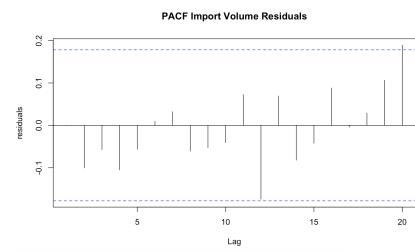
Figure 25: Residual ACF Plots



(a) Export price residual PACF



(b) Import price residual PACF



(c) Import volume residual PACF

Figure 26: Residual PACF Plots

\$JB

JB-Test (multivariate)

data: Residuals of VAR object var3_test
Chi-squared = 217.95, df = 6, p-value < 2.2e-16

\$Skewness

Skewness only (multivariate)

data: Residuals of VAR object var3_test
Chi-squared = 26.839, df = 3, p-value = 6.363e-06

\$Kurtosis

Kurtosis only (multivariate)

data: Residuals of VAR object var3_test
Chi-squared = 191.11, df = 3, p-value < 2.2e-16

Figure 27: Jarque-Bera test for normality after log-transform

```
> arch.test(var3, lags.multi = 5)
ARCH (multivariate)
```

data: Residuals of VAR object var3
Chi-squared = 297.64, df = 180, p-value = 8.013e-08

Figure 28: ARCH test

```
> arch.test(var3_test, lags.multi = 5)

ARCH (multivariate)
```

data: Residuals of VAR object var3_test
Chi-squared = 213.51, df = 180, p-value = 0.04443

Figure 29: ARCH test for logged series

```

> sarima(import_price, 3, 1, 0)
Coefficients:
      Estimate      SE t.value p.value
ar1          0.5002 0.0872  5.7363 0.0000
ar2         -0.0210 0.0984 -0.2133 0.8315
ar3         -0.2421 0.0875 -2.7667 0.0066
constant     0.7032 0.6094  1.1538 0.2509

sigma^2 estimated as 26.83207 on 121 degrees of freedom

AIC = 6.211255  AICc = 6.213921  BIC = 6.324387

> sarima(export_price, 2, 1, 1)
Coefficients:
      Estimate      SE t.value p.value
ar1         -0.2029 0.1228 -1.6519 0.1011
ar2          0.2742 0.1077  2.5464 0.0121
ma1          0.8333 0.0798 10.4405 0.0000
constant     0.7214 0.7640  0.9442 0.3469

sigma^2 estimated as 18.94985 on 121 degrees of freedom

AIC = 5.86393  AICc = 5.866597  BIC = 5.977063

> sarima(import_volume, 0, 1, 1)
Coefficients:
      Estimate      SE t.value p.value
ma1         -0.8253 0.0512 -16.1145      0
constant 138.0583 15.2451  9.0559      0

sigma^2 estimated as 878112 on 123 degrees of freedom

AIC = 16.58055  AICc = 16.58134  BIC = 16.64843

```

Figure 30: SARIMA models for the three series

References

- [1] Param S. Silvapulle and Jan Podivinsky. The effect of non-normal disturbances and conditional heteroskedasticity on multiple cointegration tests. *Journal of Statistical Computation and Simulation*, 65(1-4):173–189, 2000.